Practice Assignment 11 - Not Graded

This assignment will not be graded and is only for practice.

1 point

Let $f(x, y)$ be defined in a neighbourhood of (a, b) and $D(x, y) = f_{xx} f_{yy} - f_{xy}^2$. Suppose (a, b) is a critical point of f. Which of the following option(s) is(are) true?

- [ ] If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a point of local minimum.
- [ ] If $D(a, b) < 0$, then (a, b) is a point of local maximum.
- [ ] If $D(a, b) = 0$, then (a, b) may be a point of minimum or maximum or neither.
- [ ] If $D(a, b) < 0$, then (a, b) is a saddle point.

No, the answer is incorrect.
Score: 0
Accepted Answers:

If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a point of local minimum.
If $D(a, b) = 0$, then (a, b) may be a point of minimum or maximum or neither.
If $D(a, b) < 0$, then (a, b) is a saddle point.

1 point

Choose the correct statement(s) about $f(x, y) = x^4 + y^4$, where $D(x, y) = f_{xx} f_{yy} - f_{xy}^2$.

- [ ] $(0, 0)$ is the only critical point.
- [ ] $D(0, 0) > 0$ and $f_{xx}(0, 0) < 0$ and so $(0, 0)$ is a point of local maximum.
- [ ] $D(0, 0) = 0$.
- [ ] $(0, 0)$ is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(0, 0)$ is the only critical point.
$D(0, 0) = 0$.
$(0, 0)$ is a point of local minimum.

1 point

Let $f(x, y) = (9x^2 - 1)(1 + 4y)$ be defined on the box $-2 \le x \le 3$ and $-1 \le y \le 4$. Then which of the following options is (are) true?

- ○ $(0, 1)$ is a point of local minimum.
- ○ $(3, 4)$ is a point of absolute maximum.
- ○ -105 is the absolute minimum.

$(-2, -1)$ is a point of absolute minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(3, 4)$ is a point of absolute maximum.

1 point

Suppose $z = f(x) + yg(x)$ for some polynomial functions f and g. Which of the following options is true?

- $z_{yy} = 0$
- z_{yy} is a non-constant function of x.
- z_{yy} is a non-constant function of y.
- $z_{yy} = 1$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$z_{yy} = 0$

1 point

Define a function

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & if \ (x, y) \neq (0, 0) \\ 0 & if \ (x, y) = (0, 0) \end{cases}$$

Choose the set of correct options.

- $f_{xy}(0, 0) = -1$
- $f_{yx}(0, 0) = 1$
- $f_{xy}(0, 0) = f_{yx}(0, 0)$
- $f_x(0, 0) = f_y(0, 0)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$f_{xy}(0, 0) = -1$
$f_{yx}(0, 0) = 1$
$f_x(0, 0) = f_y(0, 0)$

1 point

Consider a function $f(x, y, z) = x^3 y + yz$. If A is the the Hessian matrix of $f(x, y, z)$ at (1,1,0), then which of the following option(s) is(are) true?

- $det(A) = 0$
-

$Rank(A) = 2$

☐ $Nullity(A) = 1$

☐ $det(A) = -6$

☐ $Rank(A) = 3$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$det(A) = -6$
$Rank(A) = 3$

1 point
Let $f: R^6 \to R$ be a polynomial function. Which of the following options is true?

○ Hessian matrix of the function is a symmetric matrix of order 6.
○ All possible mixed partial derivatives of second order have the same value at a point of R^6
○ Hessian matrix of the function is a symmetric matrix of order 5.
○ Hessian matrix of the function is a symmetric matrix of order 4.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Hessian matrix of the function is a symmetric matrix of order 6.

Let S be a rectangular box. Let x, y and z be the length of the sides of the box and volume of the box $V = xyz$. If d is the length of the diagonal of the rectangular box, then $d = \sqrt{x^2 + y^2 + z^2}$. Use this information to answer the questions 8 and 9.

1 point
If the diagonal is of length 2 units and the volume of the box is the maximum possible, then which of the following option(s) is (are) true?

☐ S is a cube.
☐ All sides are of different length.
☐ $x = \frac{2}{\sqrt{3}}$
☐ $y = \frac{1}{\sqrt{3}}$
☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
S is a cube.
$x = \frac{2}{\sqrt{3}}$

√

2

1 point

If the length of the diagonal of the box is 3 units, then which of the followings is the maximum possible volume of the box?

○

√3 3

√

cube units.

○

3√33 3

√

cube units.

○ 1 cube units.

○

33 cube units

No, the answer is incorrect.

Score: 0

Accepted Answers:

3√33 3

√

cube units.

1 point

Consider the following function

$$f(x, y, z) = x^2 y^2 + xz^2 \text{ f(x, y, z) = x2y2 + xz2}$$

Which of the following option(s) is (are) true?

☐

$(0, y, 0)$, where $y \in R$(0, y, 0), where y ∈ R are critical points.

☐

$(0, 0, z)$, where $z \in R$(0, 0, z), where z ∈ R are critical points.

☐

$(1, y, 0)$ where $y \in R$(1, y, 0) where y ∈ R are critical points.

☐

$(0, y, 1)$, where $y \in R$(0, y, 1), where y ∈ R are critical points.

☐

$(x, 0, 0)$, where $x \in R$(x, 0, 0), where x ∈ R are critical points.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, y, 0)$, where $y \in R$(0, y, 0), where y ∈ R are critical points.

$(x, 0, 0)$, where $x \in R$(x, 0, 0), where x ∈ R are critical points.

Check Answers

Your score is: 0/10